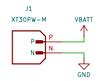
POWER MANAGMENT and SOURCES

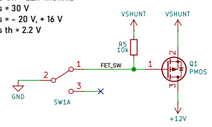
BATTERY

- 2S/3S LiPo
- 3.7V Cell voltage (nominal voltage)
- 4.2V Cell voltage (fully charged)
- 400mAh (may vary)



MAIN SWITCH

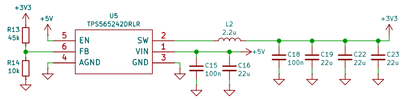
Rds On = 22.7 mOhms
Vds = 30 V
Vgs = - 20 V, + 16 V
Vgs th = 2.2 V



STEP-DOWN CONVERTER 3.3V

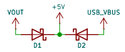
TPS565242DRLR Step down converter
Input: 3-16V
Output: 600mV - 7V
Current output: 5A
Switching freq.: 600kHz

Inductor
Inductance: 1.5uH
Max. current: 4.7A



USB POWER SWITCH

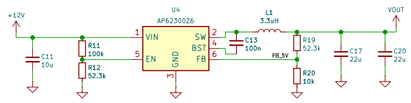
Power switching between
5V VBUS - USB
and
5V Buck converter - BATT



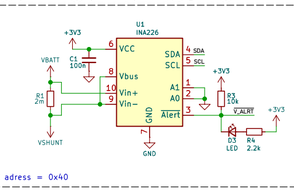
STEP-DOWN CONVERTER 5V

AP62301Z6-7 Step down converter
Input: 4.2-18V
Output: 5V
Current output: 3A
Switching freq.: 750kHz

Inductor
Inductance: 3.3uH
Max. current: 2.7A

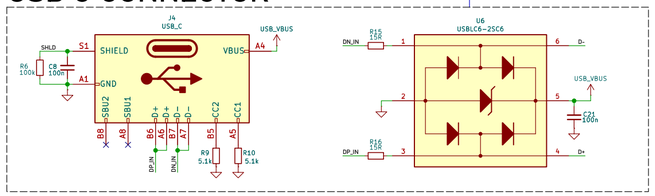


VOLTAGE READING



I2C adress = 0x40

USB C CONNECTOR



MCU and IMU

ESP32-S3-WROOM1

ESP32-S3 Datasheet

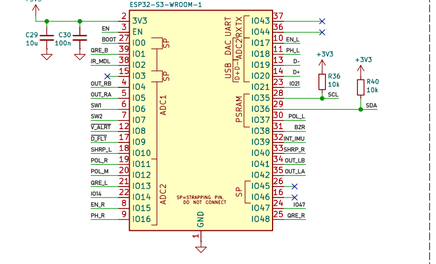
Default Configuration of Strapping Pins

Strapping Pin	Default Configuration	Bit Value
GPIO0	Pull-up	1
GPIO3	Floating	-
GPIO45	Pull-down	0
GPIO46	Pull-down	0

Typically these can be used, but you need to make sure they are not in the wrong state during boot.

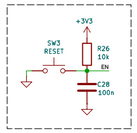
gpio.0 Boot Mode. Weak pullup during reset.
(Boot Mode 0=Boot from Flash, 1=Download)
gpio.3 JTAG Mode. Weak pull down during reset.
(JTAG Config)
gpio.45 SPI voltage. Weak pull down during reset.
(SPI Voltage 0=3.3v 1=1.8v)
gpio.46 Boot mode. Weak pull down during reset.
(Enabling/Disabling ROM Messages Print During Booting)

I2C ADDR.

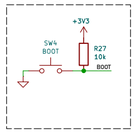


MISC

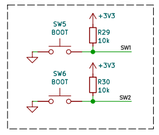
RESET



BOOT

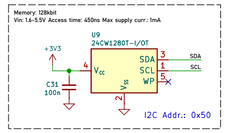


BUTTONS



EEPROM

Memory 128kbit
Vin: 1.6-5.5V Access time: 450ns Max supply curr.: 1mA



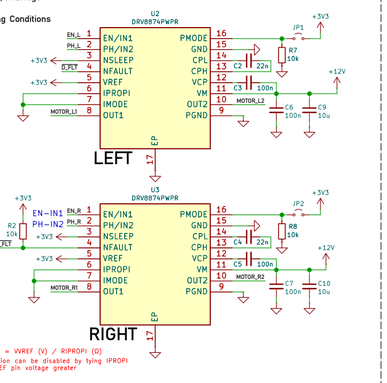
I2C Addr.: 0x50

MOTOR DRIVING

H-BRIDGE

DRV8874 - Motor driver, H-Bridge

Recommended Operating Conditions
VM = 4.5V - 37V
VIN = 0V - 5v
fPWM = 0Hz - 100kHz
Vref = 0V - 3.6V



ITRIP (A) x AIPROPI (µA/A) = VVREF (V) / RIPROPI (Ω)
The internal current regulation can be disabled by tying IPROPI to GND and setting the VREF pin voltage greater than GND

MOTORS

6-PIN JST SH-Style male connector
VCC 2.7 V to 18 V encoder operating voltage

MOTOR - LEFT

MOTOR - RIGHT

SENSORS

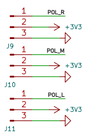
OBJECT DETECTING SENSORS

Pololu Distance Sensor with Pulse Width Output/Digital Output

Small lidar based sensor can report presnece/distance of objects up to (TOTAL MAX 200cm-Digital, 300cm - PWM)

Distance is chosen by configurating the jumpers with 0R resistors

POLOLU SENSOR MANUAL



DISTANCE MEAS.

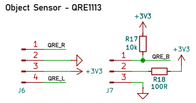
Time-of-Flight(ToF)
laser-ranging module - TF-Luna
COMMUNICATION PROTOCOL: I²C

When pin 5 is connected to ground, TF-Luna enters I2C mode, then its pin 2 is used as SDA data and pin 3 is the SCL clock sending data.

TF-LUNA MANUAL
(Better manual in -DOCS-)

LINE SENSORS

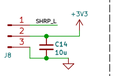
Miniature Reflective
Object Sensor - QRE1113



SIDE SENSORS

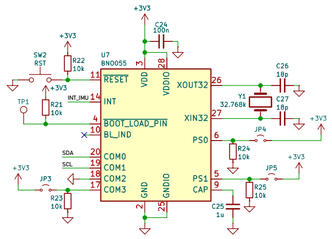
Digital distance sensor - Sharp GP2Y0D81020F

Sensor detects objects between 2 cm and 10 cm
Low when detecting an object, high otherwise.
Operating voltage 2.7V - 6.2V



9-AXIS IMU

BN0055
9 DOF Absolute IMU with
intergrated sensor fusion



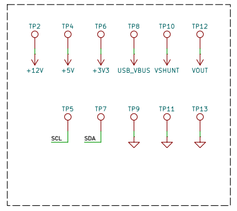
I2C Addr.: 0x29 when COM3 tied to 3V3

I2C Addr.: 0x28

PS1 (Pin 5) PS0 (Pin 6) Communication Interface
0 (GND) 0 (GND) Standard/Fast I²C (Default on most breakout boards)
0 (GND) 1 (HIGH) HID over I²C (Used for Windows plug-and-play sensor hubs)
1 (HIGH) 0 (GND) UART (Serial communication)
1 (HIGH) 1 (HIGH) Reserved (Do not use)

DRBUG and MOUNTING

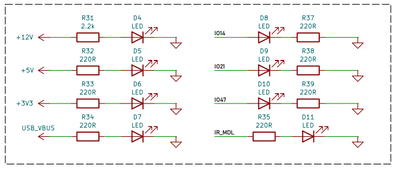
TESTING POINTS



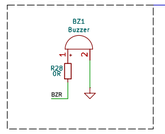
MOUNT.

M3 Padded Mounting holes

LED SIGNALISATION



BUZZER



IR MODULE

IR LED Reciver module
Operating voltage: 2.7V to 5.5V
Frequency: 38kHz

TO-DO LIST

1.
2.
3.
4.
5.
6.
7.

DEVICE NAME	I2C ADRRESS
GYROSCOPE - BMI088	0x69
ACCELEROMETER - BMI088	0x19
EEPROM - 24CW1280T	0x50
VOLTAGE READ - INU226	0x40

Description about Line Sequence and Connection

Figure 3-TF-Luna's pin outlines
The Connection of TF-Luna and Pixhawk

No.	Function	Description
1	VIN	Power Supply
2	RX(3.3V)	Receiving Pins
3	TX(3.3V)	Transmitting Pins
4	GND	Ground
5	CAN Enable Port	Low level active (V) Standard: Connect to GND or leave it open.
6	Hardware Reset	Pull up to 3.3V or leave it open. Pulse at least 2ms to reset the module.

SUMEC
MINISUMO ROBOT

SUMEC MK V.I aka SMD-V5 board
Made by bismarx-v1
CONTACT: Savva Popov, savva.popov.sp@gmail.com, +420 605 570 366
Made in Prague, Czech Republic
SPS NA PROSEKU

Sheet: /	
File: SUMEC_MK_V_Pro.kicad_sch	
Title: SUMEC_MK_V.I	
Size: A2 Date: 2025-02-27	Rev: v.1.0.0
KiCad E.D.A. 10.0.0	Id: 1/1